

Data Science Methodology

Data science combines statistics, mathematics, programming, and domain expertise to extract insights from data.

The data science lifecycle includes problem definition, data collection, data cleaning, exploratory analysis, modeling, evaluation, and deployment.

Pandas and NumPy are essential Python libraries for data manipulation and numerical computing.

Feature engineering transforms raw data into features that better represent underlying patterns for predictive models.

Statistical Analysis and Visualization

Descriptive statistics summarize data using measures like mean, median, standard deviation, and percentiles.

Hypothesis testing determines whether observed patterns are statistically significant or due to random chance.

Data visualization with matplotlib, seaborn, and plotly helps communicate insights effectively to stakeholders.

A/B testing compares two versions to determine which performs better based on statistical significance.